Vivli 2026 Cover Page

Vivli AMR Surveillance Data Challenge 2026

Final Submission Document — Cover Page

Challenge: 2026 Vivli AMR Surveillance Data Challenge
AMR Register Data Request ID: 00013346
Submission title: AMR, Life Expectancy, and Intervention Impact

Open outputs:
- Repository: https://github.com/enchill-svg/amr-vivli-2026
- Live dashboard: https://tanstack-start-app.amrlifeintel.workers.dev

Lead Applicant

Field	Entry
Name	Justice Ohene Amofa
Affiliation	Noguchi Memorial Institute for Medical Research (NMIMR)
Email	justiceoheneamofa@gmail.com
Role	Lead Applicant / primary Vivli contact

Team

Name	Affiliation	Email	Role
Justice Ohene Amofa	Noguchi Memorial Institute for Medical Research (NMIMR)	justiceoheneamofa@gmail.com	Lead Applicant
Erica Akanko	West African Centre for Cell Biology of Infectious Pathogens (WACCBIP)	eakanko15@gmail.com	Team member
Yewku Enchill-Yawson	Noguchi Memorial Institute for Medical Research (NMIMR)	enchillyawson@gmail.com	Team member
William Boateng	Noguchi Memorial Institute for Medical Research (NMIMR)	boatengwilliam72@gmail.com	Team member
Humphrey P. K. Addy	Kwame Nkrumah University of Science and Technology (KNUST)	addy.p.humphrey@gmail.com	Team member

Team type: Non-student team.
Cross-Domain eligibility: Yes — uses Global AMR R&D Hub data **and** eligible AMR Register datasets.

AMR Register datasets used in the Final Submission

Dataset	Role in this work
SOAR 201818	Primary bacterial surveillance cohort. Contributes isolates and MIC panels to the harmonized master table used for typology, trajectory, country-risk ranking, and LE/Hub analytics.
SOAR 201910	Primary bacterial surveillance cohort. Same master-table role as SOAR 201818 (organism–drug–country burden and change over time).
SOAR 207965	Primary bacterial surveillance cohort. Same master-table role; also the source of evaluability / zero-measurement handling documented in preprocessing.
Vivli/SENTRY Antifungals (2010–2024)	Primary fungal surveillance cohort in the master table (fungal typology, fungal country-year panel, fungal LE association). *Note:* stored under folder label `ATLAS_Antifungals` but this file is **SENTRY**, not ATLAS Antibiotics.

Dataset	Role in this work
ATLAS Antibiotics (2004–2024)	Integrity / evidence-gate only. *K. pneumoniae* subset used for carbapenemase detection-only Manski bounds and identifiability calibration. **Not** merged into the primary LE / country-risk master table.
PLEA Study I	Integrity / evidence-gate only. Sampling-validation reference and design-based follow-up allocator pilot (where additional MIC budget would resolve uncertainty). **Not** merged into the primary LE / country-risk master table.

Eligible Register analysis used: **Yes**.

External / supplementary datasets disclosed

Dataset	Role in this work
Global AMR R&D Hub investment / funding exports	Maps public R&D spend vs surveillance burden (Q3 funding gap); supports Global AMR R&D Hub Cross-Domain award relevance.
WHO / World Bank life expectancy and health indicators	Primary outcome and health-system context for LE association models and country explorer views.
World Bank hospital beds per capita	Health-system capacity component of public country-risk ranking (`risk_rank_core`).
ECDC ESAC-Net antimicrobial consumption	Antibiotic-consumption join for matched European country–years (Q2 driver panel; partial coverage).
WHO/UNICEF Hib3 and PCV immunization coverage	Vaccination covariates in bacterial association models where years match.
GBD 2023 Socio-demographic Index (SDI)	Dominant development covariate in fungal LE association; capacity/context for joins.
GBD LRI pathogen burden comparator	Joined for context; not used as a primary public claim series.
EUCAST Clinical Breakpoint Tables v8.1 / v10.0	Bacterial MIC → S/I/R classification reference (versioned per SOAR cohort).
CLSI antifungal breakpoints / ECVs	Fungal MIC classification / wild-type vs non-wild-type reference where breakpoints are absent.

No unused Register mirrors (e.g. SIDERO-WT, Bedaquiline/DREAM) are claimed in this Final Submission.

Project summary

Open AMR Life Expectancy Intelligence platform: a reproducible 30-stage pipeline and interactive dashboard that turn four primary Vivli AMR Register cohorts (Data Request **00013346**; **34,787** isolates; 59 countries), with ATLAS Antibiotics and PLEA Study I used for integrity calibration, into gated country-risk rankings, organism–drug typology, Hub funding–burden maps, and intervention scenarios. External joins include WHO/World Bank life expectancy, health-system indicators, GBD SDI, vaccination, ESAC-Net consumption, and Global AMR R&D Hub funding. An integrity gate publishes only estimates the data can support (`pass` / `bounds_only` / `withhold`). Findings are associational, not causal.

Software / platforms

- Analysis: Python (pandas, numpy, scipy, statsmodels, scikit-learn)
- Dashboard: TypeScript / React (TanStack Start), Cloudflare Workers
- Open access: GitHub repository and live workers.dev URL above

---

Upload checklist

- ☐ Cover Page Form completed on Vivli (use this document as the field guide)
- ☐ Final Submission PDF ≤ 5 pages including figures (cover + references excluded)

- [ ] Abstract, Objectives, Methods, Results, Impact, Tables/figures, References present
- [ ] No appendices
- [ ] Links only to open repositories (GitHub / live dashboard)
- [ ] Lead Applicant = Justice Ohene Amofa
- [ ] All Register and external datasets above disclosed

Upload via: AMR Register → Data Request **00013346** → Chat (Cover Form + Final Submission PDF).